SIADS 696 Milestone II - Project Report

Discovering Mineral Deposits

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Introduction

Mineral exploration is costly and uncertain, making it valuable to identify areas with a higher likelihood of containing economically significant mineralization before expensive drilling begins. This project investigates whether anomalous concentrations of five elements—gold (Au), silver (Ag), copper (Cu), cobalt (Co), and nickel (Ni)—can be predicted using openly available geoscientific data from Québec's SIGÉOM database.

We combine geological, geochemical, structural, and airborne magnetic data to build mineral prospectivity models. The supervised component compares Logistic Regression, XGBoost, and a multimodal CNN+MLP model that integrates magnetic imagery with tabular geological features. The unsupervised component applies PCA, UMAP, K-Means, and Gaussian Mixture Models to identify geological phenotypes and evaluate whether these clusters improve prediction performance. Together, these approaches provide a reproducible workflow for mineral prospectivity mapping using only public datasets.

Data Sources

All datasets were obtained from Québec's SIGÉOM (Système d'information géominière du Québec), an open geoscience database containing geological, geochemical, and geophysical information collected over approximately 150 years. Three primary data sources were used: airborne magnetic imagery, rock-sample geochemistry, and geological map layers containing lithology, stratigraphy, faults, and geological contacts.

The prediction targets are anomalous concentrations of gold, silver, copper, cobalt, and nickel.

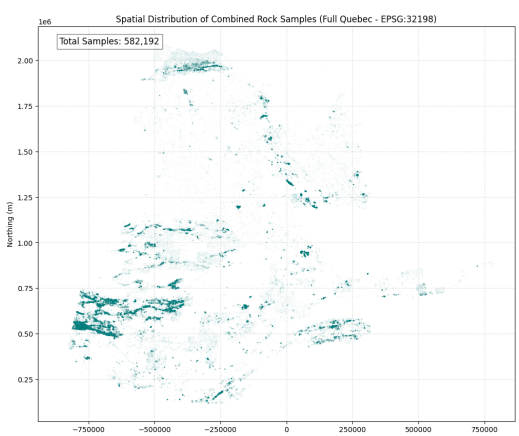

Figure 1. *Spatial distribution of all 582,192 combined rock samples across Québec (EPSG:32198). Sampling is strongly concentrated in historically explored belts — a bias quantified and addressed in validation.*

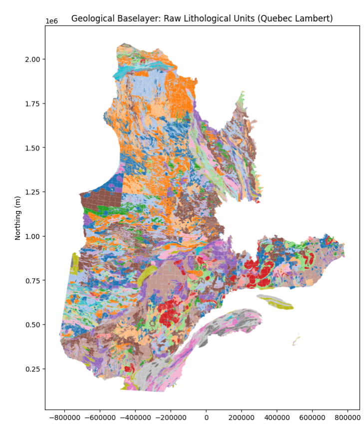

Figure 2. *Geological baselayer — raw lithological units (Québec Lambert). The polygon geology is spatially joined onto each sample to provide lithology and stratigraphy features.*

Feature Engineering

Spatial integration and structural features

Because k-means, distance-to-structure features, and tile extraction all require a metric coordinate system, every layer is first reprojected to EPSG:32198. Each rock sample is then assigned the lithology and stratigraphy of the polygon it falls within using a “within” spatial join (geopandas.sjoin); samples that fall outside the mapped polygons are labelled OUTSIDE_MAP rather than dropped . Two structural-control features are computed with a nearest-neighbour spatial join (geopandas.sjoin_nearest): the distance in metres from each sample to the nearest fault and to the nearest geological contact This programmatic pipeline reproduces, in code, a workflow previously performed by hand in QGIS, and runs over the full half-million-sample dataset in roughly 80 seconds.

Sample consolidation and outlier control

Multiple assays recorded at the same coordinates were merged into a single sample by averaging elemental concentrations while retaining the corresponding categorical and distance features. This reduced the dataset to **314,517** samples. Finally, the top **1%** of extreme values for each element were removed to reduce the influence of outliers, resulting in a final modeling dataset of **305,836** samples (Table 1).

Preprocessing stage	Rows remaining
Raw rock samples (north + south)	582,192
After removing all-missing “ghost” rows	378,513
After collapsing co-located samples	314,517
Final modeling table (after top-1% outlier trim)	305,836

Table 1. *Row counts through the preprocessing funnel, from raw SIGÉOM samples to the final modeling table.*

Anomaly label definition

For each element we convert the continuous concentration into a binary anomaly label. Rather than adopting fixed literature thresholds, we define an anomaly statistically as any value exceeding the element’s mean plus two standard deviations ($\mu + 2\sigma$), computed on the cleaned distribution. This data-driven cut-off adapts to each element’s own distribution and yields the positive prevalences in Table 2 — between roughly 2.6% (gold) and 5.6% (cobalt). The resulting imbalance (a positive rate of only 3–6%) is the defining statistical challenge of the supervised task and motivates our use of class-weighted losses and threshold-independent metrics (ROC-AUC, PR-AUC) rather than accuracy.

Element	Threshold ($\mu + 2\sigma$)	% anomalous	Positive count
Gold (AU)	421.62	2.64%	2,944
Silver (AG)	3.54	2.95%	2,856
Copper (CU)	404.43	2.91%	7,917
Cobalt (CO)	80.64	5.63%	9,940
Nickel (NI)	606.32	4.99%	9,553

Table 2. *Statistically-defined anomaly thresholds and resulting positive prevalence per element. Thresholds are in each element’s native assay units (ppb for gold, ppm for the base metals).*

Part A. Supervised Learning

Methods Description

The objective of this project was to predict the presence of gold mineralization (AU) using geological tabular data together with airborne magnetic imagery.

To ensure a comprehensive evaluation of different machine learning approaches, three diverse supervised learning model families were explored: *Logistic Regression, Extreme Gradient Boosting (XGBoost) and Convolutional Neural Network (CNN) with a Multi-Layer Perceptron (MLP)*.

The first model evaluated was **Logistic Regression**, which represents a probabilistic linear classification method that provides a simple and highly interpretable baseline against which more complex models can be compared. Since the geological dataset contains a large number of numerical and one-hot encoded categorical variables, Logistic Regression establishes the performance that can be achieved using only linear relationships among the tabular features.

The second model family explored was **XGBoost**, a tree-based ensemble learning algorithm based on gradient boosting. Unlike Logistic Regression, XGBoost can naturally capture complex nonlinear relationships and interactions between variables without requiring explicit feature engineering. Geological datasets often contain heterogeneous variables and nonlinear dependencies, making XGBoost well suited for this application.

The third model was a multimodal deep learning architecture consisting of a **Convolutional Neural Network (CNN)** combined with a **Multi-Layer Perceptron (MLP)**. The CNN processed the magnetic imagery to automatically learn spatial patterns associated with geological structures, while the MLP processed the tabular geological attributes. The latent feature representations produced by both branches were concatenated and passed through fully connected layers to generate the final prediction. By fusing these learned representations, the model can leverage both spatial and geological information simultaneously.

Overall Results

The three supervised learning models were evaluated using five-fold cross-validation to obtain reliable estimates of their predictive performance.

The target variable exhibited a strong class imbalance, with only approximately 3.2% of the samples belonging to the positive class. Consequently, model evaluation relied primarily on the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) and the Area Under the Precision-Recall Curve (PR-AUC).

Precision-Recall curves focus exclusively on the performance of the positive class by measuring the trade-off between precision and recall, making **PR-AUC** considerably more informative for highly imbalanced classification problems such as mineral prospectivity prediction.

Model Family	Feature Representation	Mean ROC-AUC	Mean PR-AUC
Logistic Regression	Tabular features	0.779 ± 0.013	0.185 ± 0.045
XGBoost	Tabular features	0.860 ± 0.009	0.329 ± 0.030
CNN + MLP	Magnetic images + tabular features	0.842 ± 0.059	0.183 ± 0.062

Table 1. *Model evaluation results*

Among the three models, XGBoost achieved the strongest overall performance, obtaining the highest ROC-AUC (0.860) and the highest PR-AUC (0.329). This indicates that the gradient-boosted decision trees were particularly effective at identifying the complex nonlinear relationships present in the geological tabular features while maintaining substantially better precision-recall performance than the other approaches. The superior PR-AUC is especially important because correctly identifying the relatively small number of positive mineralization samples is the primary objective of the prediction task.

The CNN+MLP multimodal model achieved the second-highest ROC-AUC (0.842), demonstrating that combining magnetic imagery with geological tabular information can effectively distinguish between positive and negative samples. However, despite its stronger discriminative ability compared with Logistic Regression, its PR-AUC (0.183) was nearly identical to that of the linear baseline. This suggests that the additional image information improved the model's ranking of samples overall but did not substantially improve its ability to correctly identify the scarce positive cases. Several factors may explain this outcome, including the severe class imbalance, the complexity of learning meaningful geological patterns from magnetic imagery, or the possibility that the available tabular variables already capture much of the predictive information.

Logistic Regression produced the lowest ROC-AUC (0.779), which was expected given its assumption of a linear decision boundary. Nevertheless, its performance provides a valuable baseline and demonstrates that the processed tabular features contain useful predictive information even when modeled using a relatively simple linear classifier. The modest difference between Logistic Regression and the CNN+MLP model in terms of PR-AUC further emphasizes the difficulty of improving positive-class detection in this highly imbalanced dataset.

Overall, the evaluation demonstrates that XGBoost is the most effective model for this mineral prospectivity prediction task. Its ability to model nonlinear feature interactions while remaining robust to heterogeneous tabular data resulted in the highest performance across both evaluation metrics.

Feature Importance

To better understand which types of features contributed most to the predictions of the XGBoost model, the individual feature importance scores were aggregated into four groups: *lithology one-hot encoded features, missing-value indicators, distance-related variables, and spatial coordinates*. This grouping provides a more interpretable view of how the model uses the available geological information.

Group	Importance
Lithology One-Hot-Encodings	0.844
Missing indicators	0.097
Distance	0.034
Coordinates	0.025

Table 2. *Feature groupings by importance*

The results show that the lithology one-hot encoded features dominated the model, accounting for approximately 84.4% of the total feature importance. This indicates that the geological rock classifications provide the vast majority of the predictive information used by the model.

The missing-value indicator variables contributed approximately 9.7% of the total importance. Although substantially smaller than the lithology features, this contribution is still meaningful and suggests that the pattern of missing measurements itself contains predictive information.

The contribution of the distance and coordinates variables was comparatively small. The distance-related features, including the distances to geological faults and contacts, accounted for approximately 3.4% of the total importance, while the coordinate variables (Easting and Northing) contributed only 2.5%. These results suggest that the model relies much more heavily on geological characteristics than on the absolute geographic location of each sample.

The results suggest that the model is learning meaningful geological relationships rather than depending primarily on the spatial position of the samples, increasing confidence that its predictions are based on relevant domain-specific features.

Ablation Analysis

An ablation analysis was performed to evaluate the contribution of each major feature group to the predictive performance of the XGBoost model. Starting from the best-performing model, four additional models were trained, each with one feature group removed while keeping the remaining model configuration unchanged. The evaluated feature groups were *lithology one-hot encoded variables, missing-value indicators, distance-related variables, and spatial coordinates*. Performance was compared using ROC AUC and PR AUC on the same validation split.

Model Configuration	Features Removed	ROC AUC	PR AUC	Δ ROC AUC	Δ PR AUC
Baseline	None	**0.865**	**0.329**	–	–
No Lithology OHE	Lithology one-hot encoded features	0.866	0.349	+0.000	+0.020
No Missing Indicators	`*_missing` features	0.829	0.207	-0.036	-0.122
No Distance Features	Distance to faults/contacts	0.877	0.356	+0.012	+0.027
No Coordinates	Easting, Northing	0.797	0.226	-0.068	-0.102

Table 3. *Ablation results*

The results indicate that the missing-value indicator features and the coordinate variables made the largest positive contribution to the model. Removing the missing-value indicators reduced the PR AUC from 0.329 to 0.207 and the ROC AUC from 0.865 to 0.829, representing the largest decrease in predictive performance. Similarly, removing the coordinate variables reduced the PR AUC to 0.226 and the ROC AUC to 0.797. These results suggest that both the missing-value patterns and the spatial location of each sample contain important information for predicting gold mineralization. In particular, the strong influence of the missing-value indicators implies that missing assay measurements are not completely random and instead capture meaningful characteristics of the exploration process or geological environment.

In contrast, removing the lithology one-hot encoded features did not reduce model performance. The model achieved a PR-AUC of 0.349 compared with the baseline value of 0.329, while the ROC-AUC remained essentially unchanged. Similarly, excluding the distance-related variables resulted in slightly higher ROC-AUC (0.877) and PR-AUC (0.356) than the baseline.

These findings also demonstrate that feature importance and ablation analysis provide complementary perspectives, demonstrating that highly utilized features are not always the most influential for model performance.

Sensitivity Analysis

A sensitivity analysis was conducted on the XGBoost model by varying its principal hyperparameters while keeping all other model settings fixed. The purpose of this analysis was to evaluate how sensitive the model's predictive performance was to changes in its configuration. Because the dataset is highly imbalanced (approximately 3.2% positive samples), PR AUC was used as the primary criterion for comparing configurations, while ROC AUC was used as a secondary performance measure.

Hyperparameter	Values tested	Best performing value
max_depth	3, 5, 7, 9	5
learning_rate	0.01, 0.05, 0.10	0.10
n_estimators	100, 300, 500	300
min_child_weight	1, 3, 5	3

Table 4. *Hyperparameter tuning values*

The **best performing value** configuration achieved an ROC-AUC of 0.849 and PR-AUC of 0.328.

a) **Tree Depth.** The results indicate that model performance was moderately sensitive to the choice of tree depth. Trees with a depth of five consistently produced the highest PR AUC. Shallower trees (depth three) were less capable of modeling the nonlinear relationships present in the geological features, whereas deeper trees (depths seven and nine) generally reduced PR AUC, suggesting increased overfitting to the training data.

b) **Learning Rate.** The learning rate also had a noticeable impact on performance. A value of 0.10 consistently produced the strongest results, while a learning rate of 0.01 generally yielded lower PR AUC values. The smaller learning rate likely required a larger number of trees to achieve comparable performance, whereas 0.10 provided a better balance between learning speed and predictive accuracy within the explored range.

c) **Number of Trees.** Increasing the number of estimators from 100 to 300 generally improved performance, indicating that additional boosting rounds enabled the model to learn more informative decision boundaries. However, increasing the number of trees from 300 to 500 produced little or no improvement in PR AUC, suggesting diminishing returns beyond approximately 300 estimators.

d) **Minimum Child Weight.** The parameter min_child_weight influenced the complexity of the generated trees. A value of three consistently outperformed values of one and five. Smaller values allowed the trees to become more complex and increased the risk of overfitting, while larger values constrained tree growth excessively and reduced predictive performance.

Overall, the sensitivity analysis demonstrates that the XGBoost model is reasonably robust to moderate changes in its hyperparameters. While performance varied across configurations, the differences were relatively modest, with PR AUC ranging from approximately 0.21 to 0.33. The highest-performing models consistently used moderate model complexity rather than the most aggressive parameter settings. This suggests that the predictive performance of the model depends on an appropriate balance between underfitting and overfitting, rather than simply increasing model complexity.

Tradeoff Analysis

The evaluation results highlight several important tradeoffs between model complexity, predictive performance, and feature utilization. The first and most significant tradeoff is between model complexity and predictive accuracy. Logistic Regression represents the simplest model, assuming a linear relationship between the input features and the target variable. While it is computationally efficient and easy to interpret, it achieved the lowest overall performance, with an average ROC AUC of 0.779 and a PR AUC of 0.185. In contrast, XGBoost, a more sophisticated ensemble of decision trees capable of learning complex nonlinear relationships, achieved the highest performance across both evaluation metrics, with an average ROC AUC of 0.860 and a PR AUC of 0.329. This demonstrates that the additional model complexity was justified by a substantial improvement in predictive accuracy.

A second tradeoff concerns tabular versus multimodal data integration. The CNN+MLP model combines image data with tabular geological information, making it considerably more computationally expensive to train than the purely tabular models. Despite this increased complexity, its average ROC AUC (0.842) was slightly lower than that of XGBoost, while its average PR AUC (0.183) was comparable to Logistic

Regression. These results suggest that, for the current dataset, the additional image information did not provide a meaningful improvement in identifying the minority class. This illustrates a tradeoff between incorporating richer data sources and achieving practical gains in predictive performance.

The experiments also highlight the tradeoff between model complexity and generalization. The hyperparameter tuning and ablation analyses showed that increasing model complexity does not always improve performance. For example, deeper trees and additional feature groups did not consistently increase PR AUC, and removing certain feature groups, such as the lithology one-hot encoded variables and the distance-related features, slightly improved validation performance. These findings suggest that additional features can introduce redundancy or noise, potentially reducing the model's ability to generalize to unseen data.

Finally, the project illustrates the tradeoff associated with class imbalance and evaluation metrics. Because only approximately 3.2% of the samples belong to the positive class, ROC AUC alone does not fully capture model performance. Although both XGBoost and the CNN+MLP achieved relatively high ROC AUC values, XGBoost produced a substantially higher PR AUC, indicating a much stronger ability to correctly identify positive mineralization samples while maintaining precision. Consequently, PR AUC was considered the most informative metric for model selection, as it better reflects performance on the minority class, which is the primary objective of the prediction task.

Failure analysis

To better understand the limitations of the final XGBoost model, individual prediction failures were analyzed on the held-out test set. Rather than relying solely on aggregate performance metrics, three representative misclassified records were selected and examined using SHAP (SHapley Additive exPlanations). SHAP was used to identify the features that contributed most strongly to each prediction, providing insight into the model's decision-making process.

Example 1 – High-confidence False Positive. The first example corresponds to a sample that was predicted as positive with a very high probability despite having a negative ground-truth label. The SHAP explanation showed that the prediction was primarily driven by the Easting variable together with the missing-value indicators for silver and cobalt.

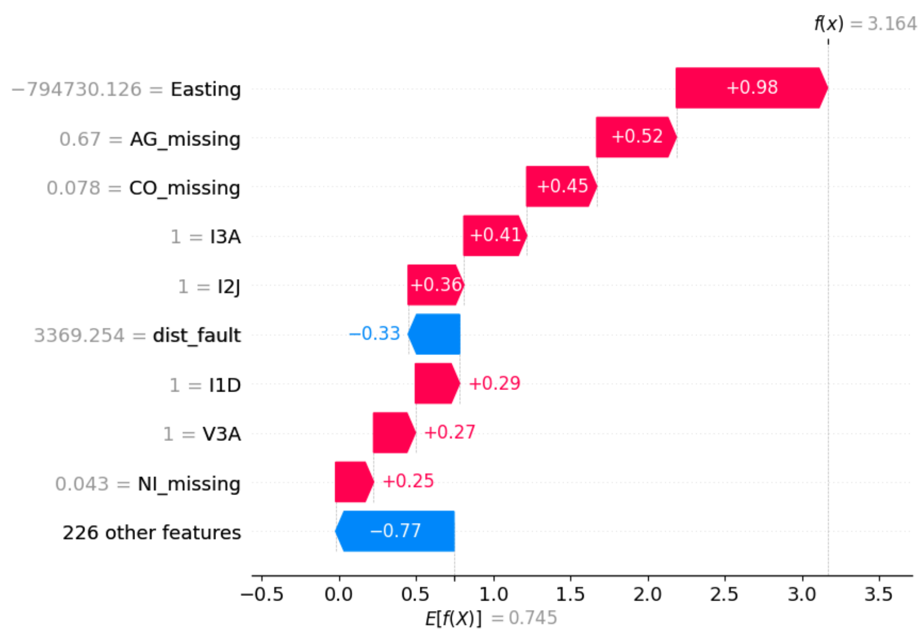

Figure 1. *Example 1 SHAP results. (Record index: 517, Probability: 0.960)*

Example 2 – High-confidence False Negative. The second example is a true mineralized sample that the model predicted as negative with high confidence. The SHAP analysis indicated that the distance to fault geological feature and the missing silver feature contributed strongly toward the negative class and outweighed the relatively small contributions from other variables.

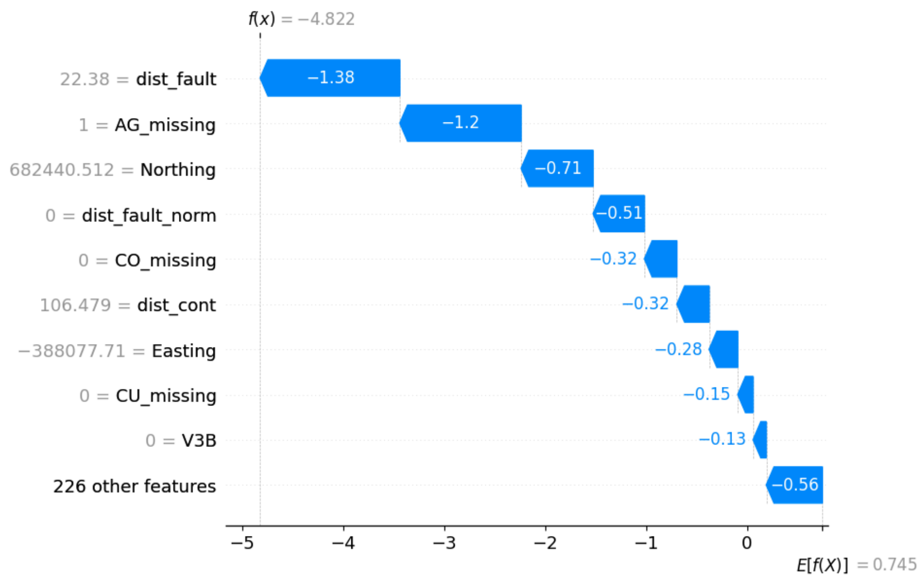

Figure 2. *Example 2 SHAP results. (Record index: 21200, Probability: 0.008)*

Example 3 – Borderline Prediction. The final example corresponds to a prediction with a probability close to the classification threshold. The SHAP explanation revealed that both positive and negative feature contributions were of similar magnitude, indicating that the model received conflicting evidence from the available features. This type of error represents an uncertain prediction rather than a confident misclassification and illustrates the difficulty of separating samples with overlapping geological characteristics.

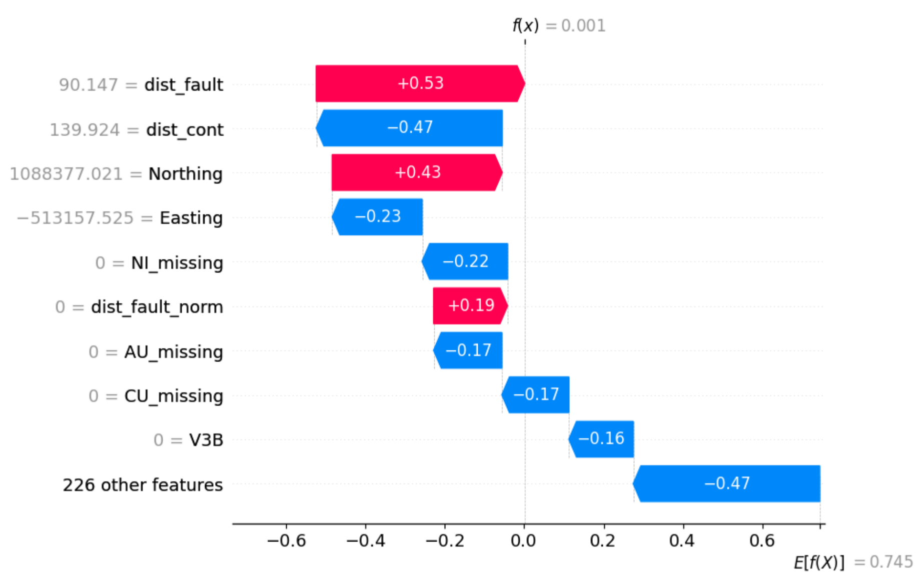

Figure 3. *Example 3 SHAP results. (Record index: 11332, Probability: 0.500)*

Overall, these examples demonstrate that the prediction failures are not caused by a single feature but instead arise from complex interactions among lithology, spatial information, and missing-value patterns.

The SHAP explanations also indicate that many failed samples possess feature combinations that closely resemble correctly classified observations from the opposite class, making them inherently difficult to distinguish. This behavior is expected in highly imbalanced geological datasets, where the decision boundary between mineralized and non-mineralized samples is often not well defined.

Several improvements could reduce these failures in future work. First, increasing the number of positive training samples or applying more advanced imbalance-handling techniques, such as focal loss or optimized class weighting, may improve the model's ability to recognize rare mineralized occurrences. Second, additional geological features, including more detailed structural, geochemical, or geophysical variables, could provide complementary information that better separates the two classes. Finally, optimizing the decision threshold rather than using the default value of 0.5 could improve the balance between precision and recall depending on the application's priorities.

Part B. Unsupervised Learning

Methods Description

The objective of the unsupervised learning component was to identify meaningful geological phenotypes within the study area without using mineral anomaly labels during clustering. Rather than directly predicting mineral occurrences, the goal was to discover natural groupings of geological samples that could reveal underlying geological structure and later evaluate whether these phenotypes provide useful information for downstream mineral prospectivity analysis.

Feature Representation

Input data. The analysis used a preprocessed dataset of 48,254 SIGEOM rock samples, reduced to 48,166 after removing 88 samples flagged as **OUTSIDE_MAP**, which lie outside Quebec's mapped bedrock coverage and would otherwise form degenerate outlier clusters.

Clustering feature set. All five binary mineral target variables (target_AU, target_AG, target_CU, target_CO, and target_NI) were excluded to prevent clustering by outcome. The feature set included distance-to-fault and distance-to-contact, 224 one-hot encoded lithology variables, 700 multi-hot stratigraphic unit variables, and log1p-transformed mineral concentrations. Continuous variables were standardized, while binary geological indicators were retained as 0/1.

Downstream prediction feature set. To avoid information leakage during downstream evaluation, all five log-transformed mineral concentration variables were excluded. The prediction models therefore used only structural and geological features (distance measures, lithology, stratigraphy) together with the learned phenotype labels, representing information realistically available before laboratory assays.

Dimensionality Reduction

Principal Component Analysis (PCA). After preprocessing, the dataset contained 928 features. PCA was applied to reduce dimensionality while retaining 95% of the total variance (190 principal components), mitigating the curse of dimensionality and improving computational efficiency for clustering.

Uniform Manifold Approximation and Projection (UMAP). UMAP was applied to the PCA representation for two purposes: a two-dimensional embedding for visualization and a 15-dimensional embedding for Gaussian Mixture Models. Compared with PCA alone, UMAP better preserves nonlinear neighbourhood relationships, allowing GMM to model more complex geological structures.

Clustering Methods

K-Means (Primary Method)

K-Means was selected as the primary clustering algorithm because of its computational efficiency, straightforward interpretation, and compatibility with the PCA feature space. As a centroid-based method, it partitions observations by minimizing within-cluster Euclidean distance, producing hard cluster assignments that can be readily incorporated into downstream XGBoost models.

The number of clusters (*k*) was tuned by evaluating values from 2 to 10 using the elbow method (inertia), Silhouette Score, and Davies–Bouldin Index. The final choice of **k = 5** achieved the highest Silhouette Score (0.191), a local minimum in the Davies–Bouldin Index (1.550), and a clear elbow in the inertia curve, as shown in figure 1. To improve robustness, **k-means++** initialization with **50 random starts** was used.

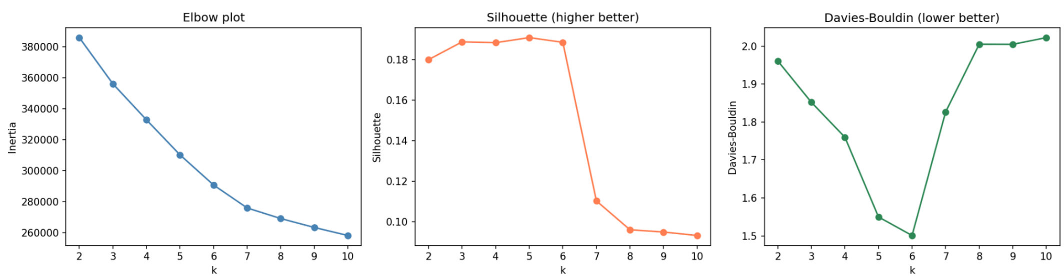

Figure 1. *K-means Selection Diagnostic Plot. A three-paneled chart displaying the Elbow Curve (Inertia vs. k), Silhouette Score Profile, and Davies-Bouldin Index across the parameter search space. It visually documents the optimization peak at K=5.*

Gaussian Mixture Model (Secondary Method)

Gaussian Mixture Models (GMMs) were selected as a complementary probabilistic approach. Unlike K-Means, GMM models the data as a mixture of multivariate Gaussian distributions and produces soft cluster memberships, making it better suited to gradual geological transitions and ellipsoidal cluster shapes.

GMM was fitted on the 15-dimensional UMAP embedding. Initially, the number of mixture components was explored using the Bayesian Information Criterion (BIC); however, as illustrated in figure 2, because BIC decreased monotonically over the search range, the final model used **five components** to match the selected K-Means solution and enable direct comparison between the two clustering families. A full covariance matrix and 20 random initializations were used to improve model stability.

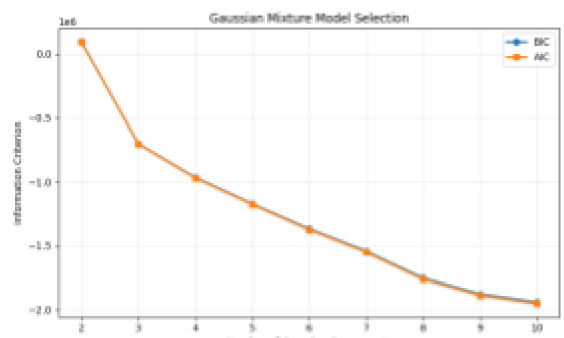

Figure 2. *Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC) across candidate numbers of Gaussian mixture components.*

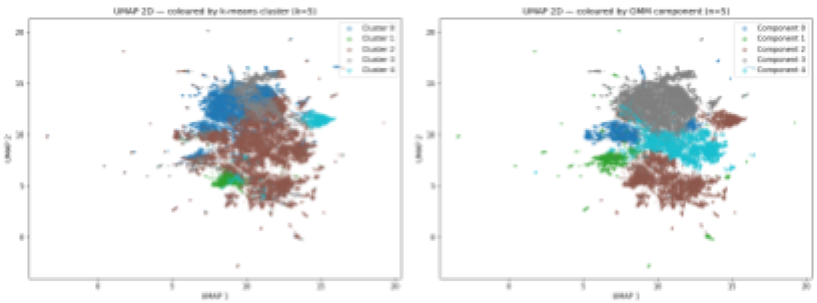

Figure 3. Two-dimensional UMAP projection of the dataset coloured by k-means cluster (left) and Gaussian Mixture Model (GMM) component (right).

Evaluation Metrics

Clustering quality was evaluated using complementary internal and external measures. For **K-Means**, model selection was based on the **Silhouette Score**, **Davies–Bouldin Index (DBI)**, and the **elbow method (inertia)**. Together, these metrics assess cluster cohesion, separation, and the trade-off between model complexity and fit.

Table 1. Summary of internal validation metrics for k=5

Metric	K-means (k=5)	GMM (n=5, matched)	Interpretation
Silhouette	0.191	—	Best across k=2–10 sweep
Davies-Bouldin	1.550	—	Local minimum in sweep
Bootstrap ARI	0.903 ± 0.105	—	STABLE (> 0.8 threshold)
Cross-algorithm ARI	0.106	(vs k-means)	Low — algorithm-dependent partition

Cluster stability was assessed using the **Adjusted Rand Index (ARI)** over 100 bootstrap resamples of 80% of the data. Agreement between K-Means and GMM was also measured using ARI to compare the consistency of the two clustering approaches.

Finally, the practical value of the learned phenotypes was evaluated using a downstream spatial holdout experiment. Three nested XGBoost models were compared for each mineral target: (A) geology-only features, (B) phenotype label only, and (C) geology plus phenotype. Performance was evaluated using **AUROC** and **PR-AUC**, with 95% confidence intervals estimated from 1,000 bootstrap samples. The difference in AUROC between Models C and A served as the primary test of whether phenotype membership provides additional predictive value beyond the original geological features.

Results

Cluster Quality and Stability

The selected **K-Means (k = 5)** solution achieved a bootstrap **ARI of 0.903 ± 0.105**, indicating excellent stability under repeated resampling. Although the maximum Silhouette Score (0.191) is modest in absolute terms, it was the highest across the full hyperparameter sweep, while the Davies–Bouldin Index reached a local minimum, supporting the selection of five clusters (Figure 2).

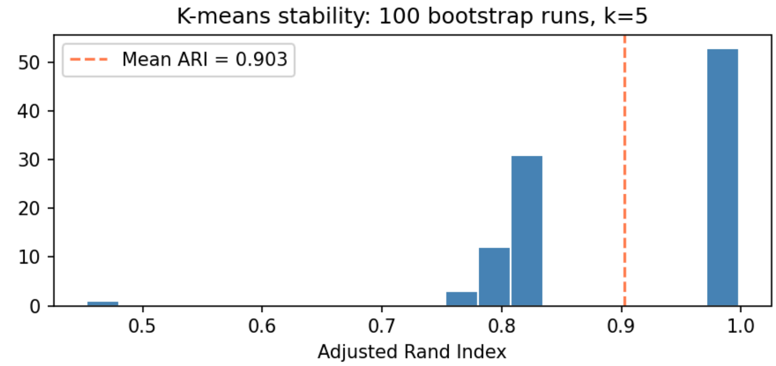

Figure 4. *Bootstrap Cluster Stability Histogram. Plots the distribution of Adjusted Rand Index (ARI) scores across 100 independent bootstrap runs on 80% data subsamples. The mean ARI is 0.9027 ± 0.1046, providing clear statistical proof that the K-means cluster structure is robust and stable against sampling noise.*

The corresponding GMM solution produced a much lower agreement with K-Means (ARI = 0.106). Rather than indicating model failure, this reflects the different assumptions underlying the two algorithms: K-Means partitions observations by centroid proximity, whereas GMM models overlapping Gaussian distributions. Consequently, each method captures different aspects of the latent geological structure.

Table 1 summarizes the quantitative comparison of the two clustering methods.

Geological Interpretation

The learned phenotypes exhibit distinct geological characteristics. As shown in **Figure 5**, Cluster 1 is enriched in gold and several base metals, suggesting a polymetallic mineralized environment, whereas Cluster 4 is characterized by elevated silver concentrations. In contrast, the two largest clusters display comparatively low precious-metal signatures, consistent with background lithologies.

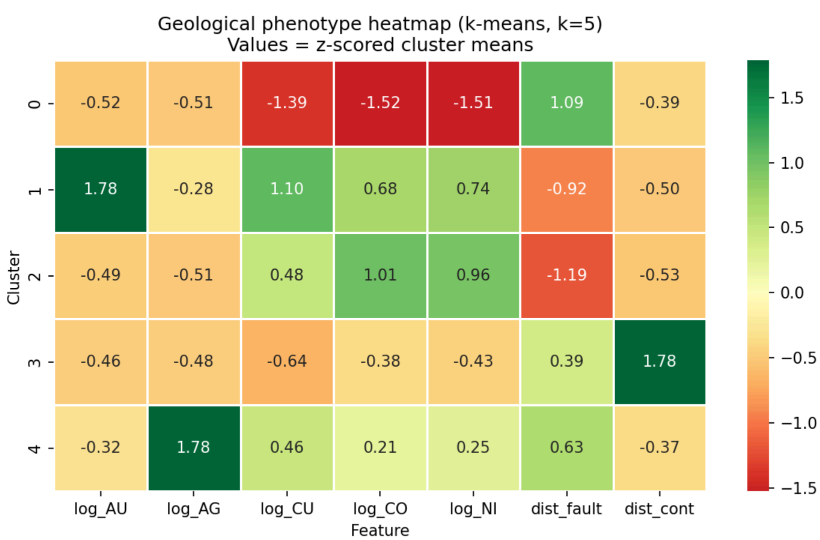

Figure 5. *Geological Phenotype Heatmap. Displays z-scored cluster means across geochemical log-minerals and fault distances. This map details the distinct physical characteristics of each phenotype, showing which clusters map to specific mineral enhancements or distinct structural baselines.*

The lithological summaries further indicate that the clusters represent characteristic combinations of geological units rather than individual rock types. The spatial distribution shown in **Figure 6** supports this interpretation, with minority clusters concentrated within the Abitibi Greenstone Belt and Grenville Province, while the two dominant clusters are distributed more broadly across Quebec. The correspondence between cluster locations and known geological provinces provides qualitative evidence that the learned phenotypes capture meaningful geological structure despite using no geographic coordinates during clustering.

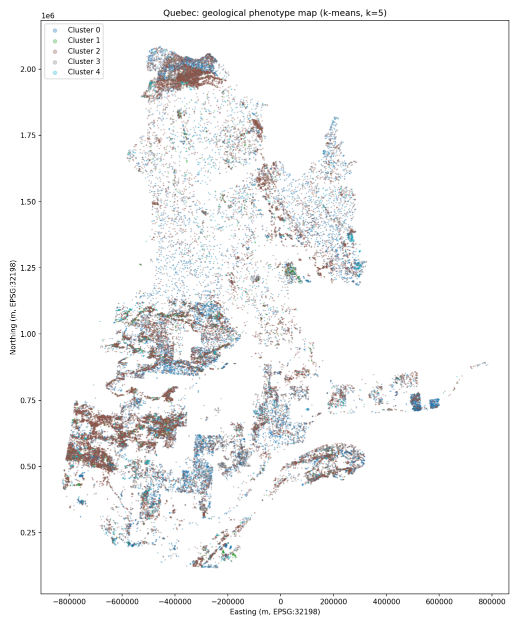

Figure 6. *Spatial distribution of clusters.*

Downstream Validation

To evaluate whether the discovered phenotypes contain information beyond the original geological variables, the cluster assignments were incorporated into downstream XGBoost models using a geographically contiguous spatial holdout. The entire preprocessing and clustering pipeline was refitted within the training region before assigning phenotype labels to the held-out test set, preventing information leakage.

Table 2: AUROC and PR-AUC for all three model variants on the spatial holdout

Mineral	Model A: Geology only	Model B: Phenotype only	Model C: Geology + Phenotype	Delta (C-A)	95% CI
AU	0.597	0.963	0.967	+0.370	[+0.131, +0.614]*
AG	0.530	0.960	0.960	+0.430	[+0.251, +0.591]*
CU	0.560	0.730	0.743	+0.183	[+0.081, +0.293]*
CO	0.618	0.662	0.708	+0.090	[+0.019, +0.169]*
NI	0.649	0.652	0.742	+0.093	[+0.015, +0.173]*

Across all five mineral targets, adding phenotype membership to the geology-only model increased AUROC, with all estimated improvements remaining positive and their 95% confidence intervals excluding zero (Table 2). The strongest and most reliable improvements were observed for copper, cobalt, and nickel, where both AUROC and PR-AUC improved consistently. Gold and silver showed larger AUROC gains; however, these results should be interpreted cautiously because the test set contained relatively few positive samples, resulting in unstable precision-recall estimates. Overall, the downstream evaluation suggests that the learned phenotypes capture complementary geological information beyond the original engineered features.

Sensitivity Analysis

Sensitivity analysis was performed by comparing neighbouring values of **k = 4, 5, and 6**. As shown in **figure 1**, clustering quality changed only modestly across these values. While *k = 4* achieved a slightly higher Silhouette Score and *k = 6* produced a marginally lower Davies–Bouldin Index, **k = 5** provided the best overall balance between cluster separation, compactness, stability, and geological interpretability.

Feature sensitivity was also explored by repeating the analysis without the stratigraphic (STRAT) variables. Removing these features resulted in only a modest reduction in clustering quality, suggesting that stratigraphic information contributes to, but is not solely responsible for, the identified geological phenotypes. Together, these analyses indicate that the selected clustering solution is robust to moderate changes in both hyperparameters and feature representation.

Discussion

Supervised Learning

One of the most surprising outcomes of this project was the difference in performance between the three model families. Although the CNN+MLP model incorporated both geological images and tabular features, the XGBoost model trained solely on the tabular data achieved the best overall performance, with an average ROC AUC of 0.860 and a PR AUC of 0.329 compared with the CNN+MLP's ROC AUC of 0.842 and PR AUC of 0.183. The ablation study further highlighted the importance of the lithology features, as removing the one-hot encoded lithology variables resulted in a substantial decrease in predictive performance. These results suggest that the engineered geological attributes contained highly informative patterns that the tree-based model was able to exploit more effectively than the deep learning architecture.

Several challenges were encountered throughout the project. The most significant was the severe class imbalance, with positive mineral occurrences representing only about 3% of the dataset. This made accuracy an unsuitable evaluation metric and required the use of ROC AUC and particularly PR AUC to properly assess model performance. The CNN+MLP model also required careful tuning of learning rate, dropout, batch size, and training epochs to reduce overfitting while maintaining generalization. Another challenge involved preprocessing the geological data, including handling missing values, generating image patches from geospatial coordinates, cleaning and one-hot encoding lithology codes, and ensuring proper alignment between the image and tabular datasets. These issues were addressed through extensive preprocessing, feature engineering, weighted loss functions, cross-validation, and systematic hyperparameter tuning.

With additional time and computational resources, the solution could be extended by combining the strengths of both deep learning and gradient boosting. Instead of training the CNN and tabular model jointly, a CNN could first be trained to extract high-level feature embeddings from the geological images. These learned image embeddings could then be concatenated with the engineered tabular features and used as the input to an XGBoost classifier. This CNN + XGBoost pipeline would leverage the CNN's ability to capture complex spatial patterns in the magnetic imagery while allowing XGBoost to model nonlinear relationships and feature interactions within the combined feature space. Given the strong performance of XGBoost on the current dataset, this hybrid approach has the potential to improve predictive accuracy while making better use of the complementary information contained in both the imagery and the geological attributes.

Unsupervised Learning

This study demonstrates that unsupervised learning can successfully identify meaningful geological phenotypes from high-dimensional geological data without using mineral anomaly labels. By combining PCA, UMAP, and clustering, the proposed workflow captured latent geological structures that were both statistically robust and geologically interpretable.

Among the two clustering methods evaluated, **K-Means** provided the best overall balance of clustering quality, stability, and interpretability. Although **Gaussian Mixture Models (GMMs)** offered greater flexibility through probabilistic cluster assignments, the resulting clusters exhibited more overlap and weaker agreement with the K-Means solution. In contrast, the K-Means clusters were more spatially coherent and easier to interpret as distinct geological environments, making them the preferred method for this dataset.

One interesting observation was that the discovered clusters did not correspond to individual lithology classes. Instead, each cluster represented a characteristic combination of lithological and geological features. This finding suggests that the learned phenotypes capture broader geological environments rather than isolated rock types, which is consistent with the gradual transitions commonly observed in natural geological systems.

Several challenges were encountered throughout the project. Selecting the optimal number of clusters required balancing multiple evaluation metrics because different criteria occasionally suggested different solutions. Additionally, careful attention was paid to preventing information leakage during downstream validation by refitting the entire preprocessing and clustering pipeline within each training split before evaluating on the held-out spatial region. These methodological improvements strengthened the reliability of the reported results.

Future work could investigate alternative clustering algorithms such as HDBSCAN or spectral clustering, explore deep representation learning methods instead of the PCA–UMAP pipeline, and evaluate the learned geological phenotypes across additional study regions. Overall, this study shows that unsupervised learning can provide meaningful intermediate geological representations that complement traditional geological features and support downstream mineral prospectivity analysis.

Ethical Considerations

Prospectivity models are decision-support tools that influence where land is disturbed; their ethical surface is therefore dominated by land use, equity, and the risk of over-reliance rather than by personal-data privacy.

Land and consent. Much of Québec's under-explored north overlaps the traditional territories of First Nations and Inuit communities. A model that directs exploration onto these lands raises questions of free, prior, and informed consent and of who benefits from any resulting discovery. Model outputs should be treated as inputs to a consultative process, not as a mandate to stake or drill.

Sampling bias and entrenchment. Historical samples cluster around previously explored regions (Figure 1), so a model trained on them may re-discover known camps while systematically under-rating genuinely under-explored terrain — reinforcing existing patterns of activity and the inequities baked into them. Our spatially-held-out validation design is partly an attempt to measure, rather than hide, this distributional shift.

Environmental footprint, both directions. Used well, focusing drilling on high-probability targets reduces unnecessary ground disturbance — a benefit the provincial survey explicitly associates with data reuse [1]. Used poorly, false positives waste capital and disturb land for nothing, while false negatives can sterilize prospective ground from future consideration. In an imbalanced setting like ours, the precision/recall trade-off is therefore an ethical choice, not merely a statistical one.

Unsupervised-specific risks. The “geological phenotypes” we derive are statistical groupings, not ground truth. k-means returns clusters whether or not real structure exists, so presenting phenotypes as definitive geological classes risks reifying artifacts. We mitigate this with stability checks and a geological face-validity step, and recommend that any operational use of cluster labels be reported together with these uncertainty measures.

Provenance and dual-use. All inputs are openly licensed government data and the outputs are regional prospectivity surfaces, so privacy and dual-use risks are minimal; the primary ethical considerations are the land-use and equity issues above.

Statement of Work

Rami Haider	Data processing and infrastructure: programmatic shapefile-to-tabular pipeline (reprojection, spatial joins, distance features, label engineering), magnetic-tile extraction and DBSCAN spatial clustering, the spatial train/test split, and multi-modal fusion design; report writing.
Rodrigo Morelli	Convolutional image branch: architecture, training, and tuning of the residual network; image augmentation; Grad-CAM / saliency analysis; image-side ROC/PR evaluation.
Shinwoo Noh	Tabular and unsupervised learning: gradient-boosted and logistic baselines with SHAP; PCA → UMAP, k-means and Gaussian-mixture clustering with internal validation; phenotype interpretation; the nested phenotype-utility evaluation.

All three authors contributed to the writing and revision of the final report.

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Appendix A — Final Feature Sets

Clustering representation (2,080 features)

- 7 continuous features: log-transformed concentrations log_AU, log_AG, log_CU, log_CO, log_NI; dist_fault; dist_cont (all z-scored).
- 2,073 one-hot features: 735 collapsed lithology (CODE_LITH) levels and 1,338 collapsed stratigraphy (STRAT) levels (rare categories < 20 occurrences merged to OTHER).
- Coordinates (Easting, Northing) are retained for mapping but excluded from the clustering distance computation.

Fusion tabular representation (235 features)

- Numeric/spatial: Easting, Northing, dist_fault, dist_cont, dist_fault_norm, dist_cont_norm.
- Missingness indicators: AU_missing, AG_missing, CU_missing, CO_missing, NI_missing.
- Multi-hot lithology columns derived from cluster-level CODE_LITH strings (delimiter-split), with columns of fewer than five occurrences dropped.

Magnetic image input

- Single 5 km × 5 km magnetic tile per spatial cluster, 200×200 px at native ≈25 m resolution, resized to 200×200 and read as a 3-channel tensor; 4,934 valid tiles.

Appendix B — Notebook Catalog

Notebook	Role
Tabular.ipynb	Shapefile → tabular preprocessing: reprojection, spatial joins, distance features, BDL/missing handling, co-located collapse, outlier trim, and $\mu+2\sigma$ label engineering. Produces geology_ml_training_data.csv.
Images (image-tiling notebook)	DBSCAN spatial clustering (eps = 500 m) and extraction of native-resolution magnetic tiles from the GeoTIFF; produces processed(500)_geology_ml_training_data.csv and the tile set.
CNN_Supervised_Model.ipynb	Multi-modal fusion model (Part A): residual image branch + tabular MLP, class-weighted training, held-out evaluation.
unsupervised.ipynb	Geological-phenotype analysis (Part B): clustering-feature engineering, PCA → UMAP, k-means and GMM, internal validation, and the nested phenotype-utility test.

Reproducibility: all source data are openly available through SIGÉOM [1]; all code is contained in the project **Github repository**